BullSO 2026 Invitational

Chemistry Lab

Bull Science Olympiad at the University of South Florida

Saturday, January 24th, 2026

Team Name:_____

Participant Names:_____

Total Score:_____**/82** **Rank:**_____

PERIODIC TABLE OF THE ELEMENTS

1 **H** 1.0079																	2 **He** 4.0026
3 **Li** 6.941	4 **Be** 9.012											5 **B** 10.811	6 **C** 12.011	7 **N** 14.007	8 **O** 16.00	9 **F** 19.00	10 **Ne** 20.179
11 **Na** 22.99	12 **Mg** 24.30											13 **Al** 26.98	14 **Si** 28.09	15 **P** 30.974	16 **S** 32.06	17 **Cl** 35.453	18 **Ar** 39.948
19 **K** 39.10	20 **Ca** 40.08	21 **Sc** 44.96	22 **Ti** 47.90	23 **V** 50.94	24 **Cr** 52.00	25 **Mn** 54.938	26 **Fe** 55.85	27 **Co** 58.93	28 **Ni** 58.69	29 **Cu** 63.55	30 **Zn** 65.39	31 **Ga** 69.72	32 **Ge** 72.59	33 **As** 74.92	34 **Se** 78.96	35 **Br** 79.90	36 **Kr** 83.80
37 **Rb** 85.47	38 **Sr** 87.62	39 **Y** 88.91	40 **Zr** 91.22	41 **Nb** 92.91	42 **Mo** 95.94	43 **Tc** (98)	44 **Ru** 101.1	45 **Rh** 102.91	46 **Pd** 106.42	47 **Ag** 107.87	48 **Cd** 112.41	49 **In** 114.82	50 **Sn** 118.71	51 **Sb** 121.75	52 **Te** 127.60	53 **I** 126.91	54 **Xe** 131.29
55 **Cs** 132.91	56 **Ba** 137.33	57 ***La** 138.91	72 **Hf** 178.49	73 **Ta** 180.95	74 **W** 183.85	75 **Re** 186.21	76 **Os** 190.2	77 **Ir** 192.2	78 **Pt** 195.08	79 **Au** 196.97	80 **Hg** 200.59	81 **Tl** 204.38	82 **Pb** 207.2	83 **Bi** 208.98	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)
87 **Fr** (223)	88 **Ra** 226.02	89 **†Ac** 227.03	104 **Rf** (261)	105 **Db** (262)	106 **Sg** (263)	107 **Bh** (262)	108 **Hs** (265)	109 **Mt** (266)	110 **§** (269)	111 **§** (272)	112 **§** (277)						

§Not yet named

*Lanthanide Series	58 **Ce** 140.12	59 **Pr** 140.91	60 **Nd** 144.24	61 **Pm** (145)	62 **Sm** 150.4	63 **Eu** 151.97	64 **Gd** 157.25	65 **Tb** 158.93	66 **Dy** 162.50	67 **Ho** 164.93	68 **Er** 167.26	69 **Tm** 168.93	70 **Yb** 173.04	71 **Lu** 174.97
†Actinide Series	90 **Th** 232.04	91 **Pa** 231.04	92 **U** 238.03	93 **Np** 237.05	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)	97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)	103 **Lr** (260)

Section I: Multiple Choice

Questions 1-6: Match reaction with the corresponding reaction type. (1pt each)

1. $Cl_2(g) + 2KBr(s) \rightarrow 2KCl(g) + Br_2(g)$ C
2. $2\ CH_3OH(g) + 3\ O_2(g) \rightarrow 2\ CO_2(g) + 4\ H_2O(l)$ A
3. $2\ CaO(s) + 2\ H_2O(l) \rightarrow 2\ Ca(OH)_2(s)$ E
4. $2\ HCl(aq) + Ca(OH)_2(aq) \rightarrow CaCl_2(aq) + H_2O(l)$ F
5. $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$ B
6. $2\ Al_2O_3(l) \rightarrow 4\ Al(l) + O_2$ D

a. Combustion

b. Metathesis

c. Single Replacement

d. Decomposition

e. Synthesis

f. Neutralization

Questions 7-10: Answer each with either **T** for **True** or **F** for **False**. (1pt each)

7. __F__ The nomenclature of $Ti_3(PO_4)_4$ is titanium (III) phosphate.
8. __F__ Ag_2SO_4 is absolutely insoluble in water.
9. __T__ Among zero, first, and second order rate laws, only the second order rate law's linearized plot has a positive slope.
10. __F__ Raising temperature increases reaction rate because it makes collisions between molecules more elastic.

Questions 11-30: Answer each of the following.

$$__\ ClO_2(g) + __\ H_2O(l) \rightarrow __\ HClO_2(aq) + __\ HClO_3(aq)$$

11. How many compounds have a coefficient of 2 when the reaction is properly balanced? (1pt)

A) 0 B) 1 C) 2 D) 3 E) 4

12. Which of the following statements are true about the reaction in Question 11? (1pt)

I. Oxygen has the same formal charge for all compounds that it's present in
II. Two reductions take place within the reaction
III. Only one element has the same formal charge for all compounds that it's present in

A) I only
B) II only
C) I and II only
D) I and III only
E) II and III only

13. How many grams of CH_4 are produced when 0.40 moles of H_2 reacts with 0.30 moles of C in the reaction below? (1pt)

$$C(s) + 2\ H_2(g) \rightarrow CH_4(g)$$

A) 0.20 g B) 1.60 g C) 3.20 g D) 4.80 g E) 6.40 g

14. How many grams of the excess reactant remain in Question 13? (1pt)

A) 0.80 g B) 1.20 g C) 2.40 g D) 3.60 g
E) None of the Above

15. What is the complete net ionic equation of the following reaction? (1pt)

$Ba(NO_3)_2(aq) + (NH_4)_2SO_4\ (aq) \rightarrow BaSO_4(aq) + 2\ NH_4NO_3(aq)$

A) $Ba^{2+}(aq) + 2\ NO_3^-(aq) \rightarrow Ba(NO_3)_2(s)$
B) $Ba^{2+}(aq) + SO_4^{2-}(aq) + NO_3^-(aq) \rightarrow BaSO_4(s) + Ba(NO_3)_2(s)$
C) $Ba^{2+}(aq) + 2\ NH_4^+(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s) + (NH_4)_2SO_4(s)$
D) $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$
E) $Ba^{2+}(aq) + NH_4^+(aq) \rightarrow BaNH_4(s)$

16. Rocky D. Bull observes the following reaction at STP:

$$2\ NO_2\ (g) + F_2\ (g) \rightarrow 2NO_2F(g)$$

Rocky's initial rate data collected is shown in the data below.

Trial	$[NO_2]$ (M)	$[F_2]$ (M)	Initial Rate (M/s)
1	0.100	0.100	2.50×10^{-4}
2	0.200	0.100	1.00×10^{-3}
3	0.100	0.200	2.50×10^{-4}
4	0.300	0.300	2.25×10^{-3}

Which of the following is the correct rate law for this reaction? (1pt)

A) Rate = $k[NO_2][F_2]$
B) Rate = $k[NO_2]^2[F_2]$
C) Rate = $k[NO_2][F_2]^2$
D) Rate = $k[NO_2]^2$
E) Rate = $k[F_2]^2$

17. What is the unit of k in the previous question? (1pt)

A) $M \cdot s^{-1}$ B) $M^{-2} \cdot s^{-1}$ C) $M^{2} \cdot s^{-1}$ D) s^{-1} E) $M^{-1} \cdot s^{-1}$

18. Which of the following statements about catalysts is **FALSE**? (1pt)

A) They lower the necessary activation energy for a reaction
B) All enzymes are biological catalysts that increase reaction rates without being permanently altered
C) A catalyst is consumed in the reaction and must be replaced as the reaction progresses
D) They have no effect on the overall energy released or absorbed
E) They increase the reaction rate by providing an alternate reaction pathway

19. Which of the following statements are true when particle size is increased? (1pt) (Tiebreaker #4)

I. Larger particles move slower, increasing the activation energy of the reaction
II. Larger particles collide with more force, increasing the reaction rate
III. Larger particles have less surface area, decreasing the reaction rate
IV. Larger particles cause the amount of effective collisions to decrease

A) I only
B) II only
C) III only
D) II and III only
E) III and IV only

20. For every 1 mole of KI in the following reaction, how many moles of product are formed? (1pt)

$Pb(NO_3)_2(aq) + 2\ KI(aq) \rightarrow ?$

A) 1.00
B) 1.25
C) 1.50
D) 2.00
E) 3.00

21. Equal volumes of 0.20 M sodium phosphate and 0.30 M calcium acetate are mixed. Which of the following statements is correct? (1pt)

A) No reaction occurs because all products are soluble
B) Calcium phosphate precipitates via a double-replacement reaction
C) Calcium acetate precipitates because acetates are insoluble
D) Sodium phosphate decomposes to form sodium oxide and oxygen
E) Sodium acetate acts as a catalyst

22. For the reaction $2A + B \rightarrow C$, the experimentally determined rate law is:

$$Rate = k[A][B]$$

Which of the following best explains this observation? (1pt) (Tiebreaker #2)

A) The reaction is elementary and occurs in a single step.
B) Only one molecule of A participates in the rate-determining step.
C) The reaction must occur through a three body collision.
D) The reaction rate is independent of temperature.
E) The derived rate law is incorrect since it does not align with the reaction.

23. Which of the following statements about catalysts is always true? (1pt)

A) A catalyst increases the rate of both the forward and reverse reactions.
B) A catalyst lowers the activation energy of the forward reaction only.
C) A catalyst lowers the activation energy of the reverse reaction only.
D) A catalyst shifts equilibrium towards the products.
E) A catalyst mainly impacts the frequency factor in the Arrhenius equation.

24. Which of the following is least likely to produce a precipitate? (1pt)

A) $AgNO_3(aq) + CaCl_2(aq) \rightarrow ?$
B) $Na_2CO_3(aq) + BaCl_2(aq) \rightarrow ?$
C) $K_2SO_4(aq) + Ca(NO_3)_2(aq) \rightarrow ?$
D) $NH_4Cl(aq) + AgNO_3(aq) \rightarrow ?$
E) $LiOH(aq) + FeCl_3(aq) \rightarrow ?$

25. A student reacts 18.0 g of Al with 80.0 g of Fe_2O_3 in the following reaction:

$$2\ Al + Fe_2O_3 \rightarrow 2\ Fe + Al_2O_3$$

Which of the following statements is true? (1pt)

A) Al is the limiting reactant and 37.3 g of Fe forms
B) Fe_2O_3 is the limiting reactant and 55.8 g of Fe forms
C) Aluminum is the limiting reactant and 55.8 g of Fe forms
D) Fe_2O_3 is the limiting reactant and 37.3 g of Fe forms
E) There is no limiting reactant

26. If only 32.0 g of Fe was produced, what's the percent yield? (1pt)

F) 82.5%
G) 85.8%
H) 88.5%
I) 91.2%
J) 95.0%

27. Describe the permanganate ion in the following reaction: (1pt)

$$MnO^-_4 + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

A) It is oxidized.
B) It is reduced.
C) It is a spectator ion.
D) It is neither oxidized nor reduced.
E) There is no permanganate here.

28. According to the rate law below, if [A] is doubled and [B] is cut in half, the rate constant changes by a factor of: (1pt)

$$Rate = k[A][B]^2$$

A) ¼ B)½ C) 1 D) 2 E) 4

29. Which of the following combinations will produce a gas upon mixing? (1pt)

A) $Na_2SO_4(aq) + BaCl_2(aq)$
B) $Na_2CO_3(aq) + HCl(aq)$
C) $AgNO_3(aq) + NaCl(aq)$
D) $KOH(aq) + HNO_3(aq)$
E) None of the above

30. Consider the following reaction below.

$$CaCO_3(s) + 2HCl\ (aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

Rocky reacts 12.0 grams of impure calcium carbonate with excess hydrochloric acid. After the reaction is done, he observes that 8.88 grams of calcium chloride were produced. What percent of the impure sample is actually calcium carbonate? (1pt)

A) 37% B) 60% C) 88% D) 74% E) 67%

Section II: Short Answer & FRQ

Questions 31- 38: Answer the following short answer questions.

31. Rocky mixes 0.50 moles of substance A in solution with 0.90 moles of substance B in solution. The reaction is A+ 2B → C. The reaction goes to completion, and the student measures the amount of product formed. How many moles of excess reactant are left over for which substance? (2pt)

____0.05 moles of A____

32. What is the correct name for $H_2C_2O_4 \cdot 2H_2O$? (2pt)

_Oxalic acid dihydrate__

33. A 7.25 g sample of $FeSO_4 \cdot _H_2O$ is heated over and over until the mass no longer changes. The final mass is 4.02 g. What is the coefficient of H_2O? (1pt)

_____7_____

34. A solution containing 0.150 mol of $Na_2CO_3(aq)$ is mixed with 500mL of 0.400M HCl(aq). Using the reaction below, how many moles of $CO_2(g)$ are produced? (1pt) (Tiebreaker #1)

$$Na_2CO_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$$

_____0.1 moles____

35. A reaction is run twice at the same temperature with identical masses of the same compounds. In Trial 1, the reactants are in large chunks. In Trial 2, the reactants are powdered. Identify which trial's reactants react faster and explain why using collision theory. (2pt)

_____Trial 2, because more surface area from powdering increases_____
____collisions and collision frequency, causing a faster reaction_____

36. Explain why mixing $AgNO_3(aq)$ and NaCl(aq) produces a reaction, but mixing $NaNO_3(aq)$ and KCl(aq) does not. (1pt)
_____ AgCl is insoluble, causing an irreversible reaction, but____
_____$NaNO_3$ and KCl are all soluble, so no reaction occurs_____

37. What charge would an ion of the following elements have? (2pt)

A) Sr +2
B) Se -2

38. In a test tube, Rocky mixes an acidic solution of tin(II) chloride ($SnCl_2$) with a solution of iron(III) chloride ($FeCl_3$) such that the resulting reaction is as follows: (2pts total)

$$SnCl_2(aq) + 2\ FeCl_3(aq) \rightarrow SnCl_4(aq) + 2\ FeCl_2(aq)$$

A) Which element is being oxidized? (1pt) Tin
B) How many electrons are transferred per atom of the element being reduced? (1pt) 1

Questions 39 - 40: Answer the following FRQs.

39. A student mixes aqueous sodium carbonate with aqueous chromium(III) nitrate. (7pts total)

a) Write the balanced molecular equation. (2pt)
$3Na_2CO_3(aq) + 2Cr(NO_3)_3(aq) \rightarrow Cr_2(CO_3)_3(s) + 6NaNO_3(aq)$
b) Write the complete ionic equation. (2pt)
$6Na^+(aq) + 3CO_3^{2-}(aq) + 2Cr^{3+}(aq) + 6NO_3^-(aq) \rightarrow$
$Cr_2(CO_3)_3(s) + 6Na^+(aq) + 6NO_3^-(aq)$

c) Write the net ionic equation. (2pt)

$2Cr^{3+}(aq) + 3CO_3^{2-}(aq) \rightarrow Cr_2(CO_3)_3(s)$

d) Explain why the reaction occurs with respect to solubility rules. (1pt)
Carbonates are insoluble other than alkali metals and ammonium, so $Cr_2(CO_3)_3$ is insoluble and forms a precipitate. This drives reaction.

40. The decomposition of a compound [X] in solution occurs by an unknown mechanism. Rocky performs several experiments and observes that doubling [X] increases the rate by a factor of four, that increasing the temperature increases the rate, and that adding a catalyst increases the rate without reaction rate but does not change the dependence on [X]. (7pts)

a) Determine the reaction order with respect to [X] and write the corresponding rate law. (2pts).
Second order, rate = $k[X]^2$

b) Using the Arrhenius equation, $k = Ae^{-Ea/RT}$, explain why the higher temperature increased the reaction rate. (1pt)
A higher t causes the exponent of e to become less negative, increasing k and therefore increasing reaction rate.

c) The Maxwell-Boltzmann distribution represents how many molecules have how much molecular kinetic energy under a constant temperature. How does the Maxwell-Boltzmann distribution shift as a result of the catalyst? (2pt) (Tiebreaker #5)
A catalyst only lowers activation energy, which does not affect kinetic energy, therefore the distribution remains unchanged.

d) What is the equation for the half-life of [X] according to the reaction order determined in part (a)? If the initial [X] is increased, what happens to half-life? (2pt) (Tiebreaker #3)
$t_{1/2} = \frac{1}{k[X]_0}$. If $[X]_0$ is increased, the half-life decreases.

Section III: Kinetics Lab Portion

Procedure:

Calibration

1. Calibrate the spectrometer such that white with a clear cuvette filled with water; dark with a black cuvette. This step will be done for you.
2. Fill a clear cuvette with stock sample of the blue dye.
3. Run a scan and select the peak of absorbance.

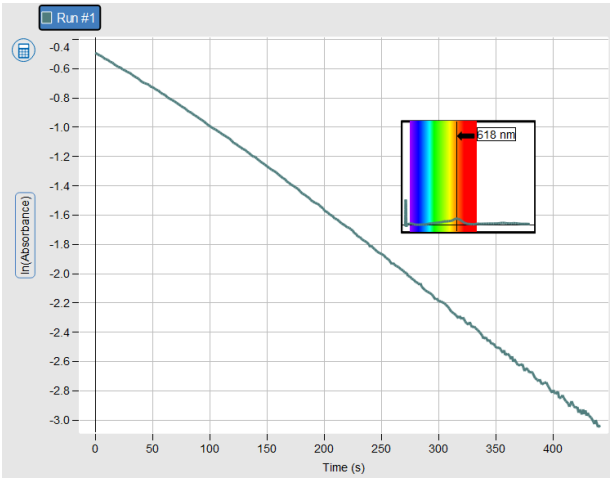
Experiment

4. Fill a beaker with 25mL of blue dye
5. Add 1mL of bleach to the solution in the beaker and mix thoroughly
6. Quickly fill one cuvette with the solution. Ensure the cuvette is filled past the cuvette tip (the narrow bottom).
7. Place the cuvette in the spectrophotometer. Ensure the arrows are facing each other.
8. Run absorbance vs. time table and wait 2-3 mins, or at least until the absorbance is past half its initial value.

Questions:

1. Draw the Integrated Rate Plot. Choose and label your axes accurately and add a title. (5pts)

   The following graph, but with a title.



2. Explain your choice of axes and which reaction order is indicated by the graph. (3pts)

   The graph plots ln(absorbance) versus time because it is most linear compared to 1/A and A. A linear ln(A) vs. time plot corresponds to the first-order integrated rate law, therefore the reaction order is 1.

3. Write the Rate Equation. What is the value of k (with unit)? (3pts)

   Rate = k[dye]. k is in s^{-1}. Actual amount of k is unneeded.

4. Justify why absorbance can be used instead of concentration when determining reaction order. (3pts)

   According to the Beer–Lambert Law, absorbance is directly proportional to concentration. Therefore, absorbance can be substituted for concentration when determining reaction order without affecting the validity of the kinetics analysis. In the integrated rate law, dividing the concentration by a constant has no effect as it cancels out from both sides due to logarithm rules.

5. Explain why the spectrophotometer was set to measure the dye's maximum wavelength instead of another wavelength. (2pts)

   The absorbance is highest at the dye's maximum wavelength so small changes in dye concentration produce the largest change in absorbance. Small wavelength shifts affect absorbance the least at the maximum wavelength and this ensures the Beer-Lambert relationship remains valid.

6. Identify one systematic error and one random error that could occur in this experiment and explain their effects on the rate constant. (4pts)

Answers may vary. Systematic errors describe a consistent repeatable error that can be fixed easily, while random errors are unpredictable and largely unavoidable. Answers must also include how they would affect the rate constant, such as whether it would increase or decrease.

7. Explain how this experiment could be modified to determine the activation energy of the reaction. (3pts)

Re-arrange the Arrhenius equation ($k = Ae^{-E_a/RT}$) into the form $ln(k) = -\frac{E_a}{R}(\frac{1}{T}) + ln(A)$. This is in slope-intercept (y = mx+b) such that y = ln k, x = 1/T, m = $-E_a/R$, and intercept is ln A. From here, experimentally finding k using the reciprocals at different temperatures would give the slope as $-E_a/R$. E_a can then be isolated to find its value.

BullSO, University of South Florida - Repaired practice edition - Answer key

Division C | Chemistry Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (2 points) Authored written substitute for unavailable source material: Water has which molecular shape?

Unit: Bonding, molecular geometry, and intermolecular forces.
Worked-case data: Why is CO_2 nonpolar while H_2O is polar?

A. Trigonal planar

B. Octahedral

C. Bent

D. Linear

Correct: C